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Failure Prediction Methodology for ESP and Operational Behavior

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Abstract

One of the most active challenges in the production life cycle is the ability to predict failure to avoid loss of production and increase deferred production. This simple question affects many areas of operation such as maintenance, supply, production, and planning. The digital transformation enables us to disrupt the conventional mode of operation and create space for such solutions.

This project aims to improve operational decision making for the logistics of maintenance planning in critical wells with ESP lifting systems of an oil field in Colombia's Upper Magdalena Valley basin. This enhancement is provided by combining Machine Learning techniques for the prediction of downhole pump failures by various agents. This project's scope is predictive failures with Analytics insight. The periodicity of failures and the impact of each of these, measured in terms of deferred production, define the criticality of the wells.

The project was developed by conducting an analysis of the state of the art in this type of prediction both within and outside the industry. Following this, historical data is collected, the data to be analyzed is selected and prepared (including sample data balancing), and the data is treated with Python. Following that, the objective is defined based on the criticality of the potential failures, and machine learning algorithms are used to determine the variables with the greatest predictive potential, and then a classification algorithm is run to predict with enough time to support decision making (hours before the event). The outcome of these processes is validated, first with information from the training data and then with hidden data.

This paper discusses the advantages of predicting failures in ESP pumps with performance greater than 60%. This is accomplished in two steps, the first of which is determining which variables contribute the most to the prediction of background failures. This analysis produced insights for field decision-making, assisting those who monitor more than 50 variables. The second step took the selected variables and delivered to maintenance planners an algorithm capable of predicting pump failures with high accuracy in enough time to perform preventive maintenance, reducing production deferrals of high-potential wells.

This project helps to increase the use of these techniques in Latin America, as well as the development of future research in the field of predictive and prescriptive analytics and the expansion of periods that lead to better tactical decision making. This solution can be extended to other fields with similar failure characteristics, allowing it to contribute to production optimization in a short period.

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Introduction

Over 90% of producer wells in the world require an artificial lift system (ALS). (VON FLATERN, 2018) mentions the importance of ALS to produce well that have no energy enough to take crude to the surface or as a way to increase initial production to improve the financial return. According to (Long Peng et al., 2021), electrical submersible pumps (ESP) are one of the most widely employed ALS, accounting for over 60% of the global oil production. This is also the ALS with the widest production range, with availability to work different casing diameters and a financial benefit up to 90% based on volumetric efficiency.

The electro submersible pumping system is composed of both subsurface and surface equipment that allow the proper functioning of the system.

The basic subsurface components of an electro submersible pumping system consist of motor, protector, intake or pump inlet and power cable, as well as some accessories such as gas separators, check and bleeder valves and centralizers among others, which are added according to the design, well conditions and operation requirements.

The standard ESP used presents several economic and technical obstacles, due to the high wear in the internal components of the pumps and their short range of operating efficiency with respect to the required production. This is the result of a low resistance of the motors to temperatures higher than 400°F generating overheating, high tendency to Downthrust and Upthrust due to inefficient designs, abrasive environments with solids and sands content higher than 6%, as well as high well deviations, designs with more rotors for higher power and high thrust and loads on the pump shaft that require more robust protectors and seals.

These obstacles result in constant maintenance services and workover operations, investment in more pump stages to meet lifting requirements and high-energy consumption, which significantly increase production costs for the company.

In general terms, the damages that can be generated in the equipment of this lifting system are:

- Excessive load for an extended period.
- Leaks in the seal or in the section of the protectors.
- Wear of the equipment by insufficient amount of fluid in movement, high temperature, corrosion and abrasion, by the conditions of the well.
- Failure due to careless or faulty installation.
- Motor controller failures,
- Imperfections in the equipment due to bad design or inadequate handling.
- Defective electrical system.
- Clogging of stages due to deposition of displaced material.
- Wear of the seals due to vibration of the pump.
- Mechanical damage of the cable during the assembly run.

The failures in the ESP system are classified based on the following parameters:

- Mechanical type failures
- Electrical type failures
- Well failures

Machine Learning Approach

ESP's are one of the most expensive equipment in artificial lift systems. That is why a failure in this kind of pumps means a huge production deferred and a big investment to restore production for operators. The large costs and the availability to get information from ESP have made some operators to focus on the development of some techniques to improve the optimization systems and the surveillance of the parameters and variables of sensors of the pump. These techniques have evolved to the use of data science and machine

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learning to find patterns to identify, diagnose and predict failures to reduce long time deferred production (Andrade Marin et al 2019).

The problem of ESP failures has been faced from different perspectives. The first one is modelling based on physics, by developing mathematical equation to describe the behavior of the equipment under different conditions. The second one is modelling based on knowledge, which are created by experts based on experience and theoretical inputs. The third one are data-driven models, which use historical data with statistics and learning techniques to find patterns of behavior (Mohannad, Lastra y Xiao 2017).

As shown in figure 1, according to the caracteristics of the problem, the available information and the desired results, we could use machine learning to interpret the extract information from the data.

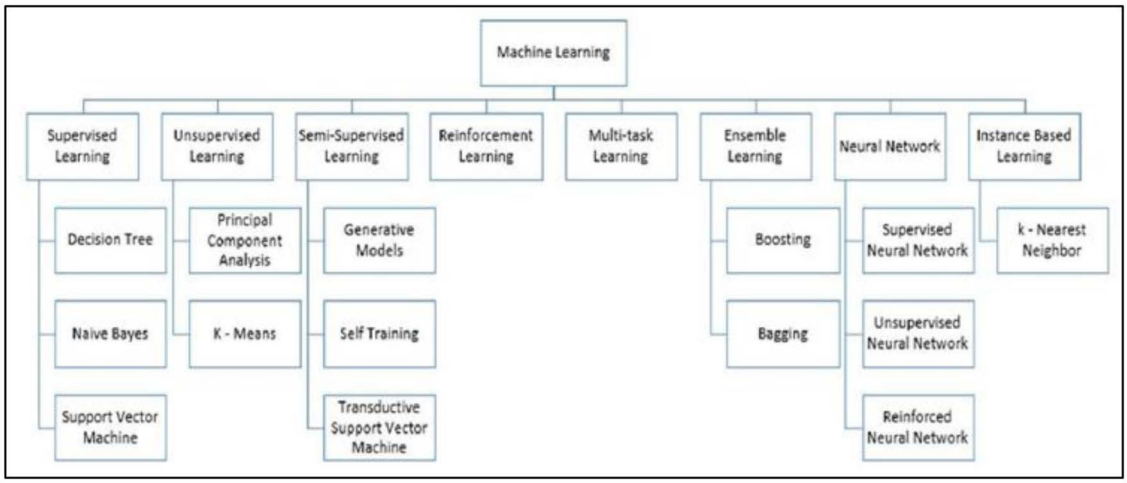

Figure 1—Machine learning algorithm classification. (Mahesh 2020)

Principal Component Analysys

Based on the review of the state of art with similar projects around the world. The algorithm selected for the analysis of the data of this study was Principal Component Analysis (PCA). This analysis is very used as a preprocessing method for data visualization and dimensionality reduction. PCA as an unsupervised dimensionality reduction technique that transforms a set of dependent variables into a set of linearly uncorrelated variables called "principal component" (Peng, Han y Sui 2021).

Also, as ESP data is usually highly correlated, using PCA allows reducing the number of parameters in a dataset by linearly combining the original parameters and creating a new space. This reduction allows evaluating a system by using only two or more principal components, instead of the original set of parameters (i.e., ten parameters), making the investigation process easier (Mohannad, Lastra y Xiao 2017).

(Mohannad, Lastra y Xiao 2017) concluded that PCA has potential to be used as a tool to identify dynamic changes in the ESP system and therefore to detect developing ESP problems. This analysis was made by building PCA models for different installations and then building a generalized PCA Model for all installations at which all normal stable conditions cluster around the origin, confirming that the dynamics between the parameters are common across all installations. PCA can also be used as unsupervised Machine Learning (ML) technique to identify hidden patterns and as an essential pre-processing technique for other ML algorithms.

Data Review

For the case of study, we had an area we 876 wells. In this area, the distribution of ALS's shows the importance of conducting analysis for ESP systems, due to a majority of the ALS (56%) as it is shown below.

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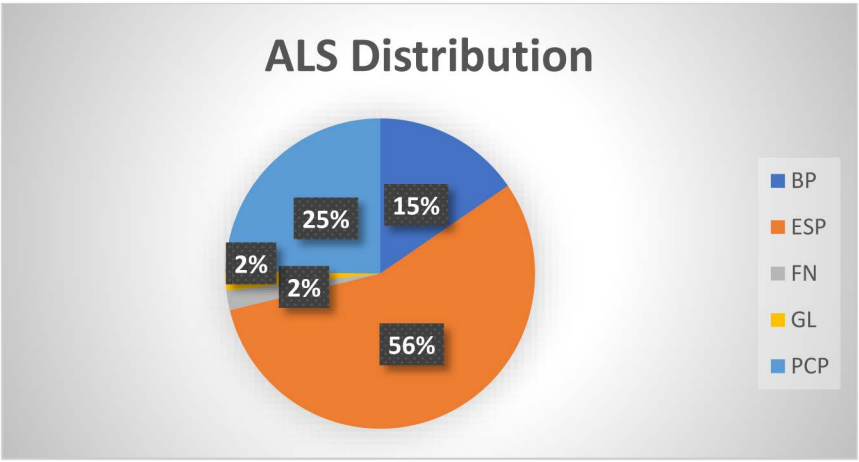

Figure 2—Distribution of ALS's in the area of study. Where BP: Beam Pump; ESP: Electro Submersible Pump; FN: Natural Flow; GL: Gas Lift; PCP= Progressive Cavity Pump.

The company identified the importance of monitoring the variables of ESP and works with devices and software that performs the function of a common interface to visualize and interact with any production data, provide supervisory control and data acquisition(SCADA)system functionality and present well-service events and histories in comparative graphs, reports, and charts to help determine best practices and operating procedures. From this system, we could extract in a (JavaScript Object Notation) JSON Format ESP downhole data as it is shown in the following table.

Table 1—Variables from downhole sensors transmitted with the system installed.

UDC	Description	UoM
ACTPWROUT	Active Power (P)	W
AMPMOTOR	Drive Amps	AMP
CT	Sensor Current	AMP
CURROTIME	Current Operating Time / 24	HRS
DESBICURR	Current imbalance	AMP
DESBIVOLT	Voltage imbalance	VOLT
FLTRVOLTS1	Output Voltage S1	VOLT
FLTRVOLTS2	Output Voltage S2	VOLT
FLTRVOLTS3	Output Voltage S3	VOLT
HZMOTOR	Motor Frequency (hz)	HZ
ITMP	Inlet Temperature	F
LASTFLT1	The last drive failure (alarm fails)	ECP
LASTOTIME	Running Time / 24 hours	HRS
MAXFREQ	Maximum frequency (setting)	HZ
MTRAMPA	Amperage Motor A	AMP
MTRAMPB	Amperage Motor B	AMP
MTRAMPC	Amperage Motor C	AMP
OUTPUTAMP	Output Amperage	AMP
OUTPUTVOLT	Output voltage	VOLT
OVERLL	Overload Limit	%
PDP	Pump Discharge Pressure	PSI
PIP	Pump Intake Pressure	PSI
PRCASXIN	Casing Pressure	PSI
PRFLOLINE	Flowline Pressure	PSI
PRTUBXIN	Tubing Pressure	PSI
PWRFACTOUT	Power factor (S)	W
RACTPWROUT	Reactive power (Q)	W
SETFREQ	Frequency setpoint	HZ
STFLOW	Flow Status	ECP
TEMPTUBXIN	Tubing Head Temperature	F
TPMOTOR	Motor Temperature	F
UNDERLL	Low Load Limit	%
VIBMOTOR	Motor Vibration	G
VOLTMOTOR	Motor Voltage (V)	VOLT
ZVIB	Z Vibration	G

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From this SCADA, the data is transmitted to the datalake of the company on cloud, where we could access it to consult it (monitoring) or to download it. The second source of information are the production and deferred reports, created by personnel of the company, which contain only structured data. The third source of reports are the fault reports, post mortem analysis, unstructured data. With this multiple sources extracted, the next step was to organize it and start the data transformation.

As part of the transformation process we transformed the data from JSON to a structured dataframe, cleaned the data, transform data from long format to wide format to facilitate the exploratory process. With this information, we could select the wells based on a three-criteria weighting. These criteria are:

1. **Number of records:** This criteria was prioritized taking into account that the most important factor to conduct this type of analysis is to have good information resources, with enough and complete data.
2. **Deferred production:** As the objective of this kind of analysis is to reduce the impact in terms of capex and deferred/lost production, one important aspect to consider choosing the wells we were going to work with was the severity of deferred production that these wells had in the last pump failures they had.
3. **Experts' opinion:** As the third criteria, we included the opinion of an expert about each of the candidates we had. This opinion was taken into account because, as in knowledge based models, there are many aspects that are hard to quantify but experts have in mind.

Exploratory Data Analysis (EDA)

The first step of constructing a model is the data analysis. According to this analysis is an approach to see what the data can tell us from the formal modeling. EDA allows analyzing the data sets and summarizing the statistical characteristics focusing on four key aspects, like, measures of central tendency, measures of spread, the shape of the distribution and the existence of outliers. (Kabita, Samal y Pramani 2019)

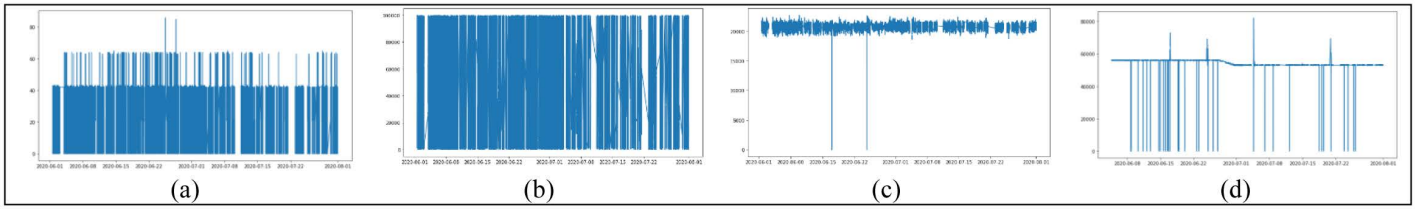

Figure 3—Exploration of some of the variables of the dataset of well XX-003 to find missing values. (a) DESBIVOLT; (b) ACTPWROUT; (c) VSDAMPB; (d) PIP.

The first exploration was the review of the data as time series to see how consistent and complete the data is in order to know if missing values are too many to affect the analysis. As we can see with the 4 graphs, the empty values of these variables are so few, compare to amount of data we have.

After this exploration, we used a library to create a complete EDA, obtaining data as quantity of values, percentage of missing values, quantity of distinct values, quantity of zeroes, percentiles, minimum, maximum, standard deviation, quartiles, median, averages, kurtosis and skew. The results of these analyses for well XX-003 are shown in figure 4.

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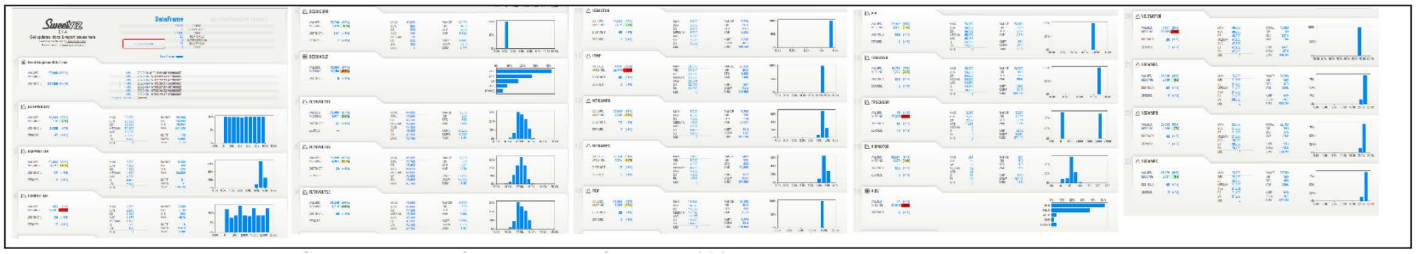

Figure 4—Exploration analysis of the variables of the dataset of well XX-003

This library also let us know the correlation between the variables with a correlation matrix. This is shown in figure 5.

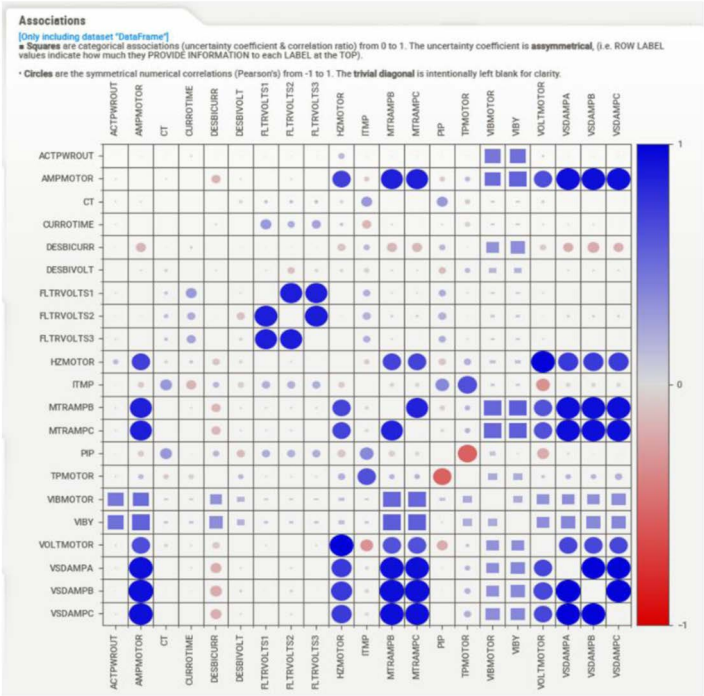

Figure 5—Correlation matrix of the variables of the dataset of well XX-003.

This matrix shows where there is a linear correlation between two variables, zero (represented by light colors and smaller size) indicates that there is no correlation between the parameters, and one and minus one (represented by dots and squares of larger size and intense colors) show respectively where the relationship is directly and inversely proportional between the well variables.

From this analysis, we could understand which of the variables had more potential to indicate a failure (for different reasons) and then use the information given by the operator to tag the data we were going to use. Additionally, from the correlation matrix we could find strong correlation between variables like the VSDAMP A, B and C. Also, correlation between variables like temperatures.

Once we confirmed the data was highly correlated, the previously established information was corroborated, where the advantage of this application was that the PCA can work with correlated data and these will not affect the performance of the principal components. With this in mind, we proceeded to prepare the data to run the model.

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Case of Study

From the exploration of the data, we found that there were about 850 wells monitored and about 53 monitored variables. Considering that this was the universe of extracted data, we proceeded to take a random sample of wells, finding that much of the information obtained was not consistent to carry out this type of study. Within the selection made, we found that the wells with the greatest amount of information had an average of 30 variables monitored consistently.

Regarding production deferrals, it was necessary to make a selection of the type of faults that were going to be the object of this study, since many of the faults that occur in these cases are not operational or are completely out of the scope of the modeling, taking as variables those obtained by downhole and surface sensors.

The data of the selected wells had duplicate information and missing information that should be completed, since this type of model should not have this missing information. Once it was verified that the information of the duplicate date records was consistent, they were eliminated. For the missing information, we used the carry forward method to complete the data for which there was no report.

With the data complete and without duplicates, the labeling of the failures was carried out in order to contrast it with the components obtained by the algorithm. With these data, we made the corresponding correlation matrices, finding the variables that best predict the behavior of the ESP in a linear way. This matrix is shown below.

	ACTPWROUT	AMPMOTOR	CURROTIME	DESBICURR	DESBIVOLT	FLTRVOLTS1	FLTRVOLTS2	FLTRVOLTS3	HZMOTOR	ITMP	MTRAMPB
ACTPWROUT	1.000000	0.017873	0.007502	0.001111	0.001702	-0.005026	-0.004860	-0.005433	0.031073	0.001342	0.019926
AMPMOTOR	0.017873	1.000000	0.005495	-0.040556	0.016602	-0.055836	-0.058890	-0.053771	0.556463	0.052332	0.909846
CURROTIME	0.007502	0.005495	1.000000	0.000007	0.023871	-0.061555	-0.069542	-0.068775	0.023180	-0.006263	0.007546
DESBICURR	0.001111	-0.040556	0.000007	1.000000	-0.003665	0.011216	0.011241	0.011711	-0.044971	0.015948	-0.047758
DESBIVOLT	0.001702	0.016602	0.023871	-0.003665	1.000000	0.020236	-0.103209	-0.058431	0.012708	0.000711	0.018997
FLTRVOLTS1	-0.005026	-0.055836	-0.061555	0.011216	0.020236	1.000000	0.978649	0.976215	-0.049718	-0.007143	-0.051652
FLTRVOLTS2	-0.004860	-0.058890	-0.069542	0.011241	-0.103209	0.978649	1.000000	0.976726	-0.051912	-0.007608	-0.054697
FLTRVOLTS3	-0.005433	-0.053771	-0.068775	0.011711	-0.058431	0.976215	0.976726	1.000000	-0.049311	-0.007543	-0.049899
HZMOTOR	0.031073	0.556463	0.023180	-0.044971	0.012708	-0.049718	-0.051912	-0.049311	1.000000	0.100309	0.578017
ITMP	0.001342	0.052332	-0.006263	0.015948	0.000711	-0.007143	-0.007608	-0.007543	0.100309	1.000000	0.054897
MTRAMPB	0.019926	0.909846	0.007546	-0.047758	0.018997	-0.051652	-0.054697	-0.049899	0.578017	0.054897	1.000000
MTRAMPC	0.020434	0.918120	0.007167	-0.032962	0.017679	-0.054509	-0.057755	-0.052698	0.568550	0.052083	0.919970
PDP	-0.002859	0.064038	-0.005606	0.016097	-0.000166	-0.005573	-0.005912	-0.005141	0.103205	0.911833	0.067036
PIP	0.001586	-0.020792	0.009214	0.021068	0.019796	-0.005799	-0.008530	-0.005587	-0.142490	0.641952	-0.031942
TPMOTOR	0.000022	0.073851	-0.005226	0.014156	-0.000385	-0.011415	-0.011801	-0.011400	0.138360	0.982145	0.077902
TPSENSOR	0.000540	0.005336	0.002877	-0.003803	-0.001223	-0.001781	-0.001694	-0.000583	0.000894	-0.753918	0.005028
VIBMOTOR	0.006954	0.053456	-0.007577	-0.040448	-0.009217	-0.019436	-0.015806	-0.017822	0.282420	-0.070884	0.078697
VIBY	-0.000477	0.004949	0.002300	-0.003558	-0.000866	0.004886	0.004934	0.005823	-0.000398	-0.737517	0.005279
VOLTMOTOR	0.031597	0.580393	0.026722	-0.033369	0.016944	-0.041043	-0.043675	-0.039250	0.948251	0.099394	0.590764
VSDAMPA	0.026808	0.640023	0.018584	-0.069874	0.024400	-0.081516	-0.085064	-0.077665	0.821252	0.081080	0.658394
VSDAMPB	0.026853	0.638998	0.017424	-0.081663	0.022213	-0.076979	-0.080238	-0.073153	0.828212	0.080342	0.662031
VSDAMPC	0.027376	0.639413	0.018364	-0.034374	0.023533	-0.077427	-0.080931	-0.073545	0.826147	0.081738	0.658571
falla	-0.030259	-0.573522	-0.025624	-0.126143	-0.016163	0.041666	0.044445	0.040020	-0.916087	-0.103209	-0.583719

	PDP	PIP	TPMOTOR	TPSENSOR	VIBMOTOR	VIBY	VOLTMOTOR	VSDAMPA	VSDAMPB	VSDAMPC	falla
ACTPWROUT	-0.002859	0.001586	0.000022	0.000540	0.006954	-0.000477	0.031597	0.026808	0.026853	0.027376	-0.030259
AMPMOTOR	0.064038	-0.020792	0.073851	0.005336	0.053456	0.004949	0.580393	0.640023	0.638998	0.639413	-0.573522
CURROTIME	-0.005606	0.009214	-0.005226	0.002877	-0.007577	0.002300	0.026722	0.018584	0.017424	0.018364	-0.025624
DESBICURR	0.016097	0.021068	0.014156	-0.003803	-0.040448	-0.003558	-0.033369	-0.069874	-0.081663	-0.034374	-0.126143
DESBIVOLT	-0.000166	0.019796	-0.000385	-0.001223	-0.009217	-0.000866	0.016944	0.024400	0.022213	0.023533	-0.016163
FLTRVOLTS1	-0.005573	-0.005799	-0.011415	-0.001781	-0.019436	0.004886	-0.041043	-0.081516	-0.076979	-0.077427	0.041666
FLTRVOLTS2	-0.005912	-0.008530	-0.011801	-0.001694	-0.015806	0.004934	-0.043675	-0.085064	-0.080238	-0.080931	0.044445
FLTRVOLTS3	-0.005141	-0.005587	-0.011400	-0.000583	-0.017822	0.005823	-0.039250	-0.077665	-0.073153	-0.073545	0.040020
HZMOTOR	0.103205	-0.142490	0.138360	0.000894	0.282420	-0.000398	0.948251	0.821252	0.828212	0.826147	-0.916087
ITMP	0.911833	0.641952	0.982145	-0.753918	-0.070884	-0.737517	0.099394	0.081080	0.080342	0.081738	-0.103209
MTRAMPB	0.067036	-0.031942	0.077902	0.005028	0.078697	0.005279	0.590764	0.658394	0.662031	0.658571	-0.583719
MTRAMPC	0.064518	-0.026104	0.074602	0.007251	0.059642	0.006949	0.591482	0.652173	0.651769	0.653638	-0.585302
PDP	1.000000	0.569066	0.908121	-0.594765	-0.055232	-0.575456	0.110120	0.095807	0.093642	0.095998	-0.112070
PIP	0.569066	1.000000	0.598336	-0.489016	-0.308196	-0.478228	-0.083604	-0.026964	-0.038182	-0.032167	0.050794
TPMOTOR	0.908121	0.598336	1.000000	-0.713606	-0.043802	-0.696640	0.137674	0.112314	0.111566	0.113273	-0.139513
TPSENSOR	-0.594765	-0.489016	-0.713606	1.000000	0.148202	0.997674	0.000627	0.001039	0.001337	0.000482	-0.000714
VIBMOTOR	-0.055232	-0.308196	-0.043802	0.148202	1.000000	0.148376	0.101484	0.081413	0.111001	0.088046	-0.087040
VIBY	-0.575456	-0.478228	-0.696640	0.997674	0.148376	1.000000	-0.000280	0.000326	0.000494	-0.000296	0.000291
VOLTMOTOR	0.110120	-0.083604	0.137674	0.000627	0.101484	-0.000280	1.000000	0.857963	0.848303	0.859740	-0.969985
VSDAMPA	0.095807	-0.026964	0.112314	0.001039	0.081413	0.000326	0.857963	1.000000	0.991985	0.994040	-0.847793
VSDAMPB	0.093642	-0.038182	0.111566	0.001337	0.111001	0.000494	0.848303	0.991985	1.000000	0.991221	-0.838685
VSDAMPC	0.095998	-0.032167	0.113273	0.000482	0.088046	-0.000296	0.859740	0.994040	0.991221	1.000000	-0.850446
falla	-0.112070	0.050794	-0.139513	-0.000714	-0.087040	0.000291	-0.969985	-0.847793	-0.838685	-0.850446	1.000000

Figure 6—Correlation matrix of the variables of the dataset of well XX-003.

This information is our first approach to identify the variables that could help us to explain linearly the behavior of the failures. With this information, we could start action plans to set some alerts for field operators.

From this point on, the PCA analysis begins, for which a standardization and balancing of the data is performed. In this way we ensure that later in the execution of a classification algorithm to predict failures in real time for future approaches, there will be no problems due to the low number of records with failures or the weight that a single variable can add to the main components or to the prediction algorithm directly.

PCA with two components.

The first PCA is performed with only two components, which seek to explain the variance of 22 variables that enter this model. The selection of the number of components is done with the objective of keeping

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the model as simple as possible in order to generate simple explanations to the variance in the increase of principal components. For well XX-003 the results are shown below.

Table 2—Results of PCA with two Principal Components for well XX-003

	PC1	PC2	falla
0	-2.342846	0.421452	False
1	-2.582702	0.021029	False
2	-2.264909	-0.617189	False
3	-2.263540	0.135952	False
4	-2.391566	-0.056444	False
...	...	...	...
32449	4.557043	0.572616	True
32450	4.661948	0.364825	True
32451	4.561247	0.571902	True
32452	4.661948	0.364825	True
32453	-1.559609	0.094263	True

32454 rows × 3 columns

These PC1 and PC2 results explain the variance present in the monitored variables of this well. PC1 explains about 70% of this variance and PC2 explains about an additional 12%. In this way, and only in two dimensions, the more than 20 initial variables with which the problem was proposed could be explained. This variance explanation values is shown below.

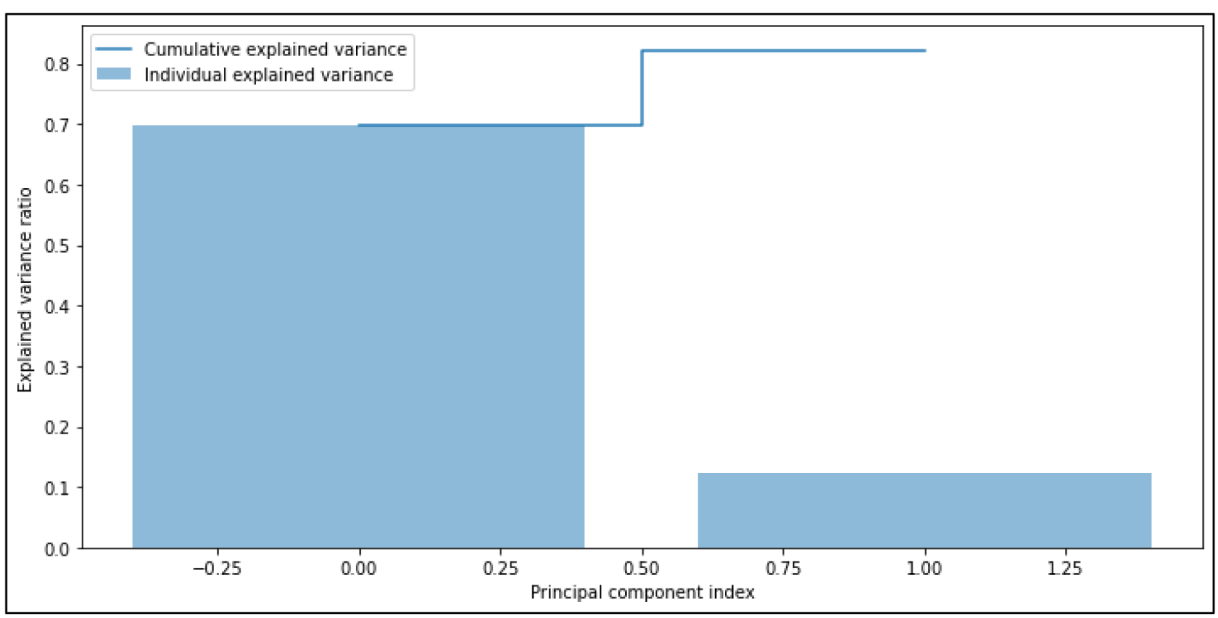

Figure 7—Explained variance ratio for PCA with two components.

In the following graph a two-dimensional visual representation can be given to make it easier to understand how to interpret these components. The yellow dots shown represent the false values, in which no pump failure occurs. The red dots represent the moments of pump failure.

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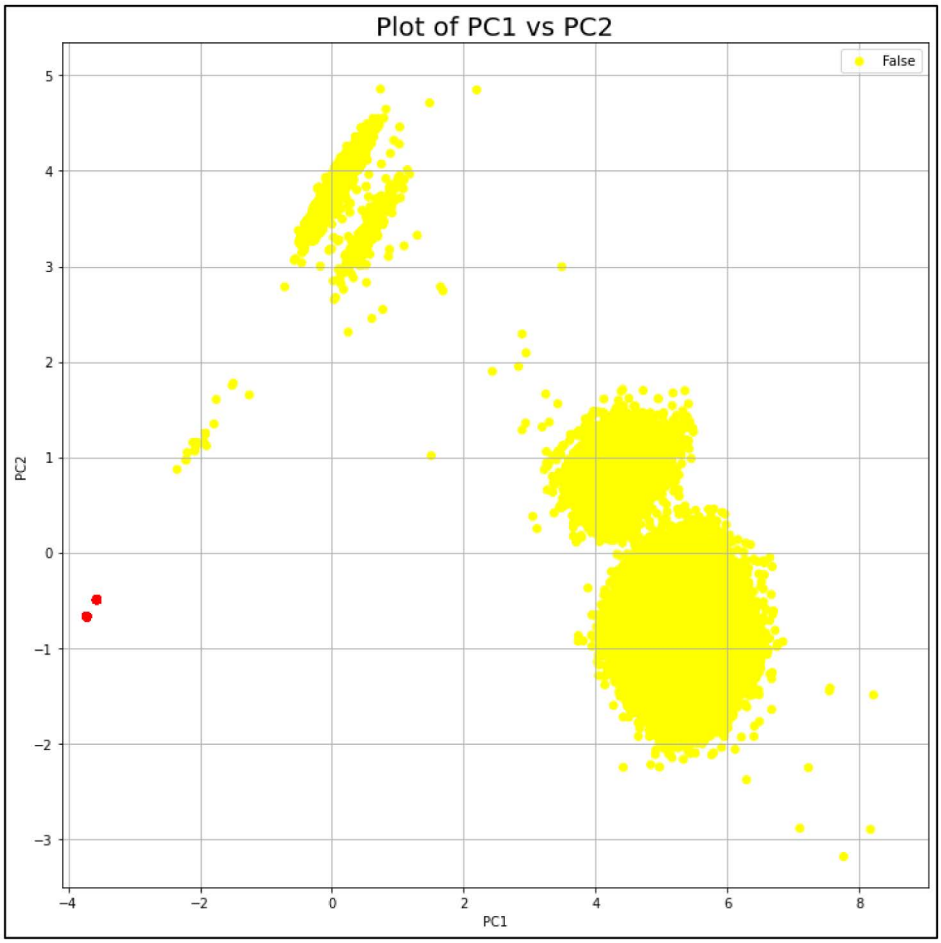

Figure 8—PC1 vs PC2 for well XX-003.

In the left zone of the graph are the red dots that show the recorded pump failures. In the zone where component 1 registers values lower than -3 is where the failures for the well in question are located. For the yellow points, the components show zones that could be classified into new categories, due to the groups that are identified.

PCA with three components.

For the resulting components, the contribution value of each of the studied variables is obtained. The table with the values of this contribution is presented below. The values marked in red show the variables with the greatest contribution to PC1. On the other hand, the variables highlighted in blue are those with the greatest contribution to PC2.

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Table 3—Variables contribution to components 1 and 2

	PC1	PC2
ACTPWROUT	0.156299	0.221144
AMPMOTOR	0.236451	0.191258
CT	0.210923	0.303971
CURROTIME	0.182392	0.003675
DESBICURR	0.22112	0.160341
DESBIVOLT	0.197235	0.144243
FLTRVOLTS1	0.227029	0.255182
FLTRVOLTS2	0.227034	0.254904
FLTRVOLTS3	0.227015	0.255164
HZMOTOR	0.235755	0.175216
ITMP	0.222926	0.274248
MTRAMPB	0.236532	0.190514
MTRAMPC	0.236564	0.190125
PIP	0.17895	0.371913
TPMOTOR	0.235763	0.176791
VIBMOTOR	0.188602	0.198182
VIBY	0.187864	0.231571
VOLTMOTOR	0.236271	0.176458
VSDAMPA	0.236651	0.190124
VSDAMPB	0.236627	0.190224
VSDAMPC	0.236666	0.189962

Subsequently, a PCA with three components was performed to improve the explanation of the variance obtained with the two-component model. For this case, the categories of the data are kept as false and true to the pump failure condition. The results of this model are presented below.

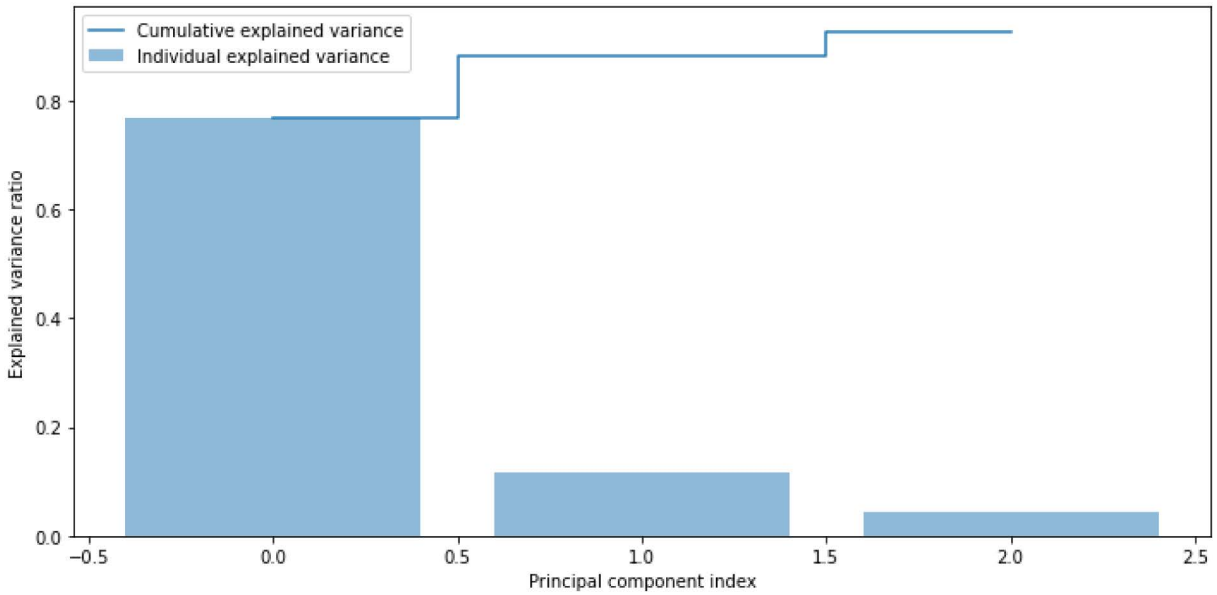

Figure 9—Explained variance ratio for PCA with three components.

This graph shows that the incremental contribution generated by the three-component analysis is 5%. This value, despite being incremental, is a low value compared to the contribution made by the first two components.

Figure 10 shows how some separated zones are generated in (a) and (b) for the faults. This allows to create zones where pump failures can be assumed to exist. In (c) these zones are not so clear.

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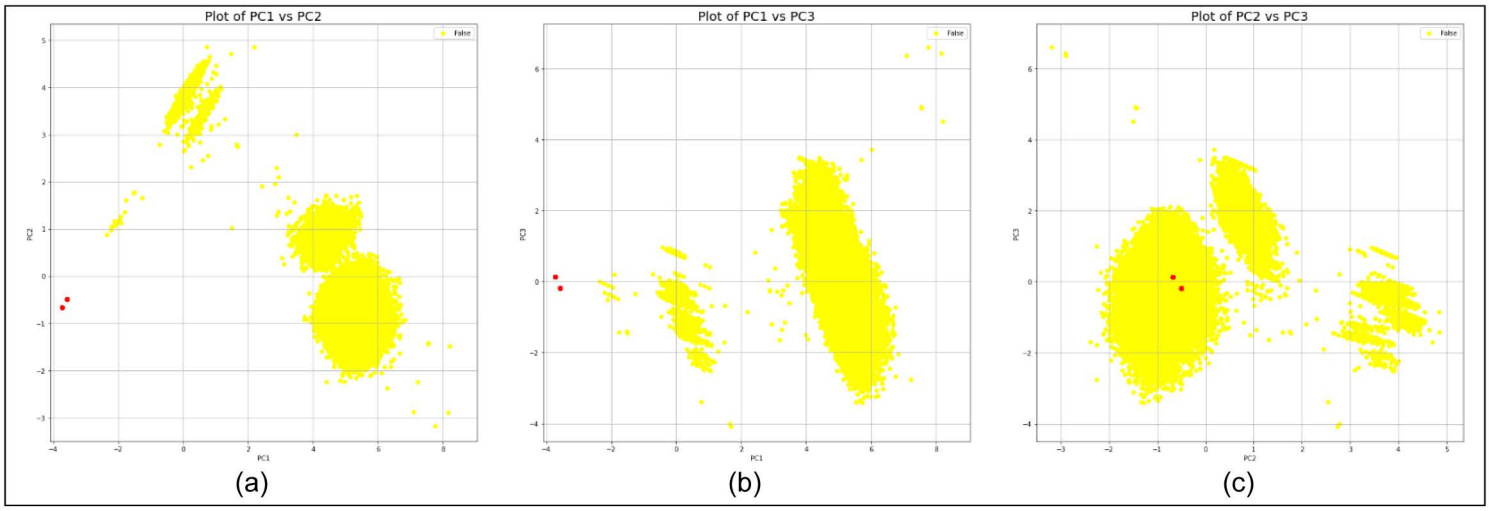

Figure 10—PC1 vs PC2 vs PC3 for well XX-003. (a) PC1 vs PC2; (b) PC1 vs PC3; (c) PC2 vs PC3.

PCA with label data for three components.

In order to determine if the presence of some groups within these graphs have some relation with some conditions before or after the failure, a different labeling is made, where some periods of time in normal flow, a reduced time before the failure, the time of the failure and the logs after the reactivation of the well after the failure are identified with colors. The results obtained from this test are shown below.

Depending on the value identified, the following conditions are generated:

Group 1: At the time of failure.
Group 2: Before the failure.
Group 3: After the fault.
Group 4: Long before the failure, more than two hours before the failure.

The PCA was run getting the following results:

Table 4—Components 1, 2 and 3 for labeled data

	PC1	PC2	PC3	failure
0	3.164771	0.573279	-0.234355	4
1	1.205478	-0.135053	-0.961528	3
2	-3.156130	-0.213431	-0.569590	3
3	2.898280	0.630585	-0.391735	4
4	2.788127	0.424730	-0.941799	4
...	...	...	...	...
90635	2.975458	0.319587	0.264285	3
90636	1.387543	-0.069560	-0.322326	3
90637	-3.000585	-0.548325	-0.895879	3
90638	-3.064488	-0.391831	-0.718012	3
90639	-3.117130	-0.291715	-0.646379	3

90640 rows × 4 columns

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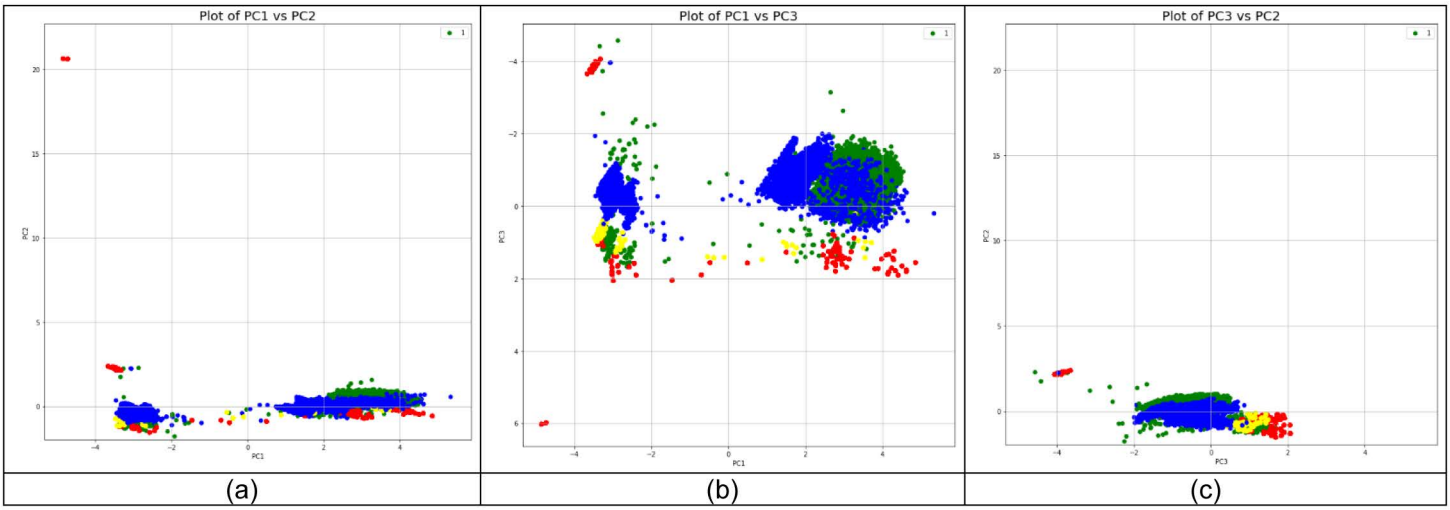

Figure 11—PC1 vs PC2 vs PC3 with label data for well XX-003. (a) PC1 vs PC2; (b) PC1 vs PC3; (c) PC2 vs PC3. Where red: Group 1; yellow: Group 2; blue: Group 3; Green: Group 4.

With this results we can find different zones in (b) and (c) for the different moments of the operation and failure of the pump.

Conclusions and Results

It is recommended to the operating company to review the process of generation and storage of information, because the error of the JSON files may be increasing, leaving the information in the data lake useless.

The data sample provided by the company allows understanding the behavior of the ESP pumps.

It is recommended to the company to validate the storage processes from the source to the datalake, in order to ensure the integrity of the data for further analysis.

The involvement of the user area is important in data analysis as it allows correlating the data with the operational reality, bringing data science closer to the operation.

Data exploration through the visualization of parameters in time series and the association matrix allows to understand the relationships of the parameters in an analytical way.

The collinearity matrix allows an understanding of the possible relationships that exist between the data and the objective function in a visual and quick way.

Given the classification generated with the PCA, it is possible to identify three components in which the pump failure is predicted a few hours in advance and the failure moments are identified.

The model is configured so that it can be implemented in production in such a way that it could identify the prediction of failures in real time.

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